Merge Sort Complexity Analysis

Topic: TimeComplexity File: p5.pdf

PROBLEM

Sort an array by dividing it into halves, sorting each half, and merging the sorted halves.

RECURRENCE

T(n) = 2T(n / 2) + O(n): two recursive halves plus a linear-time merge.

ANALYSIS

There are about log2(n) levels of splitting. At each level, merging processes O(n) total elements.

EXAMPLE

[8, 3, 5, 1] becomes [3, 8] and [1, 5], then merges to [1, 3, 5, 8].

COMPLEXITY

Best, average, and worst time: O(n log n). Typical array implementation uses O(n) auxiliary space.

TAKEAWAY

The Master Theorem gives O(n log n) for this recurrence.